

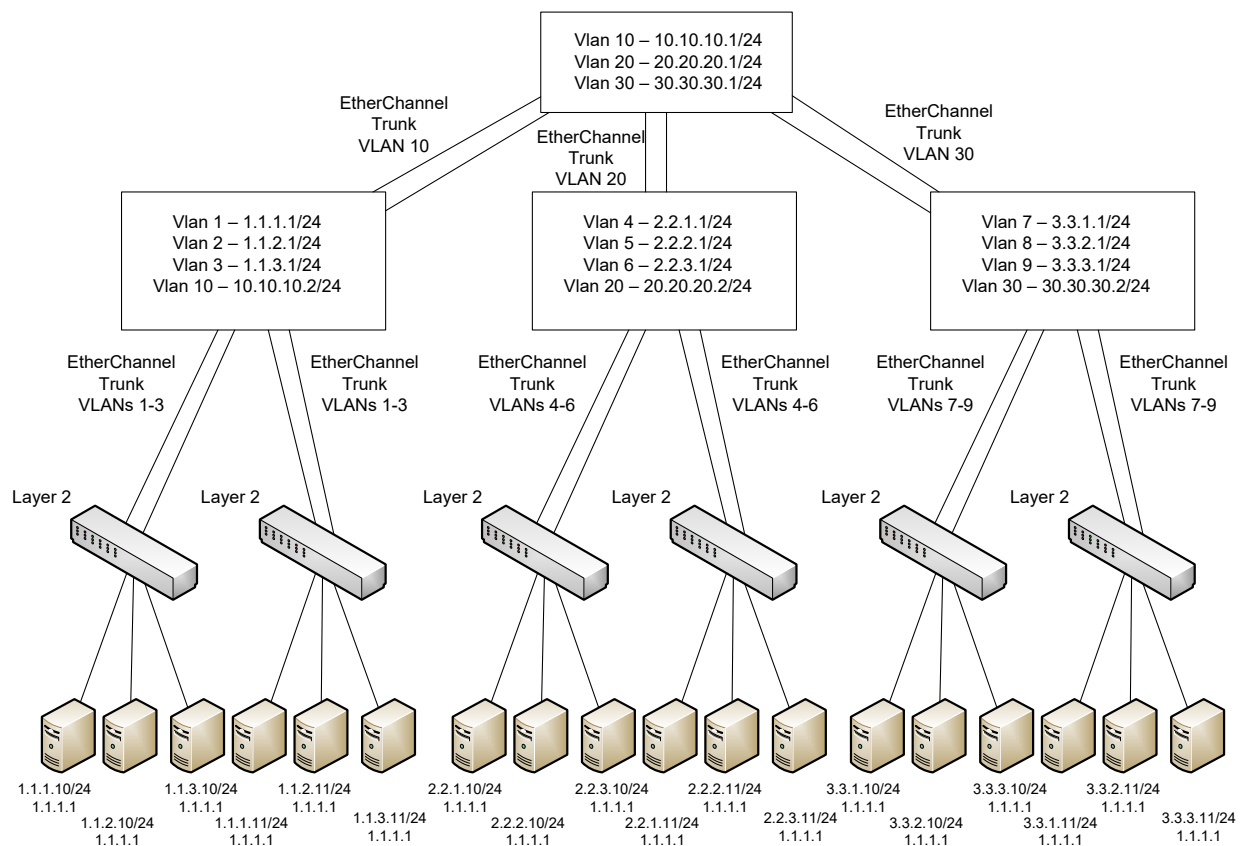
Lab 11

3-Tier Architecture

Objective:

The purpose of this lab is for the student to understand, create and implement a network core using 3-tier architecture.

Diagram:



Components:

18 – PC's with an Ethernet NIC

4 – Layer 3 Switches

6 – Layer 2 Switches

Procedure:

1. Connect all PC's together and put their static IP addresses, subnet masks and gateways on them.
2. Label the Distribution Layer 3 switches VLAN IP addresses below

Distribution Layer 3 Switch (1)

VLAN	IP
1	1.1.1.1/24
2	1.1.2.1/24
3	1.1.3.1/24
10	10.10.10.2/24

Distribution Layer 3 Switch (2)

VLAN	IP
4	2.2.1.1/24
5	2.2.2.1/24
6	2.2.3.1/24
20	20.20.20.2/24

Distribution Layer 3 Switch (3)

VLAN	IP
7	3.3.1.1/24
8	3.3.2.1/24
9	3.3.3.1/24
30	30.30.30.2/24

3. Label the routing tables for the Layer 3 switch and routers below.

Dist. Switch 1	Dist. Switch 2	Dist. Switch 3
C-1.1.1.0/24	C-2.2.1.0/24	C-3.3.1.0/24
C-1.1.2.0/24	C-2.2.2.0/24	C-3.3.2.0/24
C-1.1.3.0/24	C-2.2.3.0/24	C-3.3.3.0/24
C-10.10.10.0/24	C-20.20.20.0/24	C-30.30.30.0/24
D-2.2.0.0/16 -> 10.10.10.1	D-1.1.0.0/16 -> 20.20.20.1	D-1.1.0.0/16 -> 30.30.30.1
D-3.3.0/16 -> 10.10.10.1	D-3.3.0.0/16 -> 20.20.20.1	D-2.2.0.0/16 -> 30.30.30.1

4. Label the Core Layer 3 switches VLAN IP addresses below

Core Layer 3 Switch

VLAN	IP
10	10.10.10.1/24
20	20.20.20.1/24
30	30.30.30.1/24

5. Label the routing table for the Core Layer 3 switch below.

Core Layer 3 Switch

C-10.10.10.0/24
C-20.20.20.0/24
C-30.30.30.0/24
D-1.1.0.0/16 -> 10.10.10.2
D-2.2.0.0/16 -> 20.20.20.2
D-3.3.0.0/16 -> 30.30.30.2

6. Label the Layer 2 Switches ports and what vlans they are in below.

Access Switch 1	Access Switch 2	Access Switch 3
Port 1 – Vlan 1	Port 1 – Vlan 1	Port 1 – Vlan 4
Port 2 – Vlan 2	Port 2 – Vlan 2	Port 2 – Vlan 5
Port 3 – Vlan 3	Port 3 – Vlan 3	Port 3 – Vlan 6
Port 23&24 – Trunk (Vlan 1-3)	Port 23&24 – Trunk (Vlan 1-3)	Port 23&24 – Trunk (Vlan 4-6)

Access Switch 4	Access Switch 5	Access Switch 6
Port 1 – Vlan 4	Port 1 – Vlan 7	Port 1 – Vlan 7
Port 2 – Vlan 5	Port 2 – Vlan 8	Port 2 – Vlan 8
Port 3 – Vlan 6	Port 3 – Vlan 9	Port 3 – Vlan 9
Port 23&24 – Trunk (Vlan 4-6)	Port 23&24 – Trunk (Vlan 7-9)	Port 23&24 – Trunk (Vlan 7-9)

7. Label the Layer 2 Switches EtherChannel ports and what ports are included in the bundle.

Acc. Sw1	Acc. Sw2	Acc. Sw3	Acc. Sw4	Acc. Sw5	Acc. Sw6
Port 1 – Fa0/23-24	Port 1 – Fa0/23-24	Port 1 – Fa0/23-24	Port 1 – Fa0/23-24	Port 1 – Fa0/23-24	Port 1 – Fa0/23-24

8. Label the Distribution Switches trunk ports and what vlans are allowed.

Distribution Switch 1	Distribution Switch 2	Distribution Switch 3
Port 21-22 – Trunk (Vlan 1-3)	Port 21-22 – Trunk (Vlan 4-6)	Port 21-22 – Trunk (Vlan 7-9)
Port 23-24 – Trunk (Vlan 1-3)	Port 23-24 – Trunk (Vlan 4-6)	Port 23-24 – Trunk (Vlan 7-9)
Port 1-2 – Trunk (Vlan 10)	Port 10-11 – Trunk (Vlan 20)	Port 15-16 – Trunk (Vlan 30)

9. Label the Distribution Switches EtherChannel ports and what ports are included in the bundle.

Distribution Switch 1	Distribution Switch 2	Distribution Switch 3
Channel 1 – Fa0/1-2	Channel 1 – Fa0/10-11	Channel 1 – Fa0/15-16
Channel 2 – Fa0/21-22	Channel 2 – Fa0/21-22	Channel 2 – Fa0/21-22
Channel 3 – Fa0/23-24	Channel 3 – Fa0/23-24	Channel 3 – Fa0/23-24

10. Label the Core Switches trunk ports and what vlans are allowed.

Core Switch
Channel 1 – Ports Fa0/1-2 – Vlan 10
Channel 2 – Ports Fa0/10-11 – Vlan 20
Channel 3 – Ports Fa0/15-16 – Vlan 30

11. Connect all of the switches together as shown in the diagram
- What type of cable did you use? I used straight through because they're newer switches, but it should probably be crossover
12. Set all switches to have a vtp domain of Netcom
- What command did you use? Vtp domain Netcom

13. Set all Layer 3 switches to use vtp mode server
 - a. What command did you use? Vtp mode server

14. Set all Layer 2 switches to use vtp mode client
 - a. What command did you use? Vtp mode client

15. On the core switch create the VLANs and put the proper IP's on them

16. On the core switch place the ports into their proper ether channels
 - a. What commands did you use to place one pair of ports into an EtherChannel?
Interface port-channel1
Switchport trunk allowed vlan 10
Switchport trunk encapsulation dot1q
Switchport mode trunk
Interface range fa0/1-2
Channel-group 1 mode on

17. On the core switch place the EtherChannel ports into trunk mode and prune off all unused vlans.
 - a. What commands did you use to place a portchannel into trunk mode and prune the vlans?
These commands are included in #16

18. On the core switch enable routing and EIGRP with an AS of 1. Distribute out all connected networks.
 - a. What commands did you enter?
Ip routing
Router eigrp 1
No auto-summary
Network 10.10.10.0 0.0.0.255
Network 20.20.20.0 0.0.0.255

Network 30.30.30.0 0.0.0.255

19. On the Distribution switches place the ports into their proper ether channels
20. On the Distribution switches place the EtherChannel ports into trunk mode and prune off all unused vlans
21. On the Distribution switches create the VLANs and put the proper IP's on them
22. On the Distribution switches enable routing and EIGRP with an AS of 1. Distribute out all connected networks. Place all interfaces into passive mode except for the VLAN's that connect to the Core switch
 - a. What commands did you enter to place the ports into passive mode and core connected vlans into active mode?
Router eigrp 1
No auto-summary
Passive-int def
No pass vlan 10
Network 10.10.10.0 0.0.0.255
Network 1.1.1.0 0.0.0.255
Network 1.1.2.0 0.0.0.255
Network 1.1.3.0 0.0.0.255
23. Place the Layer 2 switch ports into their proper ether channels
24. Place the Layer 2 switch EtherChannel ports into trunk mode and prune off all unused vlans
25. Place the Layer 2 switch ports into the proper vlans. Put the ports in access mode and portfast mode

- a. What commands did you enter to place a single port in vlan 2, access and portfast modes?

Int fa0/2

Spanning-tree portfast

Switchport mode access

Switchport access vlan 2

26. Verify all the PC's can ping each other

27. Can this design allow a single vlan to be on every switch (such as vlan 5) if needed?

It's possible, but some things would have to be changed.

- a. What would have to happen?

VLAN pruning would have to be removed or it would have to include vlan 5

28. Where would an internet connection be connected to?

The core switch/router would be connected to the internet

- a. What components would be needed?

A firewall and a connection to an ISP internet connection

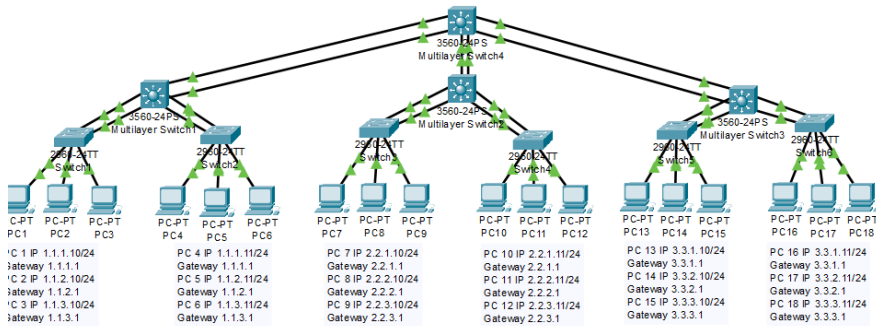
29. Where would a server farm be connected to?

Probably the distribution layer so that it could more easily be accessed.

- a. What components would be needed?

A dedicated distribution layer switch with its own set of access switches and a connection between the new distribution switch and the core.

30. Create the lab diagram below (PCs, router and switch connections) in Packet Tracer or Visio. **Label the IP addresses, subnet masks, and gateways** used by each device. **Label the routing tables next to the switch and router** and the **switches vlan IP addresses and port vlan information**. Label Port Channels and members.



Core Switch VLAN Table

VLANs	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30 active
Vlan10	10.10.10.1 YES manual up
Vlan20	20.20.20.1 YES manual up
Vlan30	30.30.30.1 YES manual up
Port-channel 1 - VLAN 10 Fa0/12	
Port-channel 2 - VLAN 20 Fa0/10,11	
Port-channel 3 - VLAN 30 Fa0/15,16	

Dist Switch 1 VLAN Table

VLANs	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30 active
Vlan1	1.1.1.1 YES manual up
Vlan2	1.1.2.1 YES manual up
Vlan3	1.1.3.1 YES manual up
Vlan10	10.10.10.2 YES manual up
Port-channel 1 - VLAN 10 Fa0/12	
Port-channel 2 - VLAN 1-3 Fa0/21,22	
Port-channel 3 - VLAN 1-3 30 Fa0/23,24	

Dist Switch 2 VLAN Table

VLANs	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30 active
Vlan4	2.2.1.1 YES manual up
Vlan5	2.2.2.1 YES manual up
Vlan6	2.2.3.1 YES manual up
Vlan20	20.20.20.2 YES manual up
Port-channel 1 - VLAN 20 Fa0/10,11	
Port-channel 2 - VLAN 4-6 Fa0/21,22	
Port-channel 3 - VLAN 4-6 30 Fa0/23,24	

Dist Switch 3 VLAN Table

VLANs	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30 active
Vlan7	3.3.1.1 YES manual up
Vlan8	3.3.2.1 YES manual up
Vlan9	3.3.3.1 YES manual up
Vlan30	30.30.30.2 YES manual up
Port-channel 1 - VLAN 30 Fa0/15,16	
Port-channel 2 - VLAN 7-9 Fa0/21,22	
Port-channel 3 - VLAN 7-9 30 Fa0/23,24	

Core Switch Routing Table

1.0.0.0/24 is subnetted, 3 subnets
D 1.1.1.0 [90/51225600] via 10.10.10.2, 00:03:46, Vlan10
D 1.1.2.0 [90/51225600] via 10.10.10.2, 00:03:46, Vlan10
D 1.1.3.0 [90/51225600] via 10.10.10.2, 00:03:46, Vlan10
D 2.0.0.0/24 is subnetted, 3 subnets
D 2.2.1.0 [90/51225600] via 20.20.20.2, 00:03:46, Vlan20
D 2.2.2.0 [90/51225600] via 20.20.20.2, 00:03:46, Vlan20
D 2.2.3.0 [90/51225600] via 20.20.20.2, 00:03:46, Vlan20
D 3.0.0.0/24 is subnetted, 3 subnets
D 3.3.1.0 [90/51225600] via 30.30.30.2, 00:03:47, Vlan30
D 3.3.2.0 [90/51225600] via 30.30.30.2, 00:03:47, Vlan30
D 3.3.3.0 [90/51225600] via 30.30.30.2, 00:03:47, Vlan30
D 10.0.0.0/24 is subnetted, 1 subnets
C 10.10.10.0 is directly connected, Vlan10
C 20.0.0.0/24 is subnetted, 1 subnets
C 20.20.20.0 is directly connected, Vlan20
C 30.0.0.0/24 is subnetted, 1 subnets
C 30.30.30.0 is directly connected, Vlan30

Dist Switch 1 Routing Table

1.0.0.0/24 is subnetted, 3 subnets
C 1.1.1.0 is directly connected, Vlan1
C 1.1.2.0 is directly connected, Vlan2
C 1.1.3.0 is directly connected, Vlan3
D 2.0.0.0/24 is subnetted, 3 subnets
D 2.2.1.0 [90/76825600] via 10.10.10.1, 00:01:07, Vlan10
D 2.2.2.0 [90/76825600] via 10.10.10.1, 00:01:07, Vlan10
D 2.2.3.0 [90/76825600] via 10.10.10.1, 00:01:07, Vlan10
D 3.0.0.0/24 is subnetted, 3 subnets
D 3.3.1.0 [90/76825600] via 10.10.10.1, 00:01:07, Vlan10
D 3.3.2.0 [90/76825600] via 10.10.10.1, 00:01:07, Vlan10
D 3.3.3.0 [90/76825600] via 10.10.10.1, 00:01:07, Vlan10
C 10.0.0.0/24 is subnetted, 1 subnets
C 10.10.10.0 is directly connected, Vlan10
D 20.0.0.0/24 is subnetted, 1 subnets
D 20.20.20.0 [90/51225600] via 10.10.10.1, 00:01:07, Vlan10
D 30.0.0.0/24 is subnetted, 1 subnets
D 30.30.30.0 [90/51225600] via 10.10.10.1, 00:01:07, Vlan10

Dist Switch 2 Routing Table

1.0.0.0/24 is subnetted, 3 subnets
D 1.1.1.0 [90/76825600] via 20.20.20.1, 00:01:36, Vlan20
D 1.1.2.0 [90/76825600] via 20.20.20.1, 00:01:36, Vlan20
D 1.1.3.0 [90/76825600] via 20.20.20.1, 00:01:36, Vlan20
D 2.0.0.0/24 is subnetted, 3 subnets
C 2.2.1.0 is directly connected, Vlan4
C 2.2.2.0 is directly connected, Vlan5
C 2.2.3.0 is directly connected, Vlan6
D 3.0.0.0/24 is subnetted, 3 subnets
D 3.3.1.0 [90/76825600] via 20.20.20.1, 00:01:36, Vlan20
D 3.3.2.0 [90/76825600] via 20.20.20.1, 00:01:36, Vlan20
D 3.3.3.0 [90/76825600] via 20.20.20.1, 00:01:36, Vlan20
D 10.0.0.0/24 is subnetted, 1 subnets
D 10.10.10.0 [90/51225600] via 20.20.20.1, 00:01:36, Vlan20
D 20.0.0.0/24 is subnetted, 1 subnets
C 20.20.20.0 is directly connected, Vlan20
D 30.0.0.0/24 is subnetted, 1 subnets
D 30.30.30.0 [90/51225600] via 20.20.20.1, 00:01:36, Vlan20

Dist Switch 3 Routing Table

1.0.0.0/24 is subnetted, 3 subnets
D 1.1.1.0 [90/76825600] via 30.30.30.1, 00:02:35, Vlan30
D 1.1.2.0 [90/76825600] via 30.30.30.1, 00:02:35, Vlan30
D 1.1.3.0 [90/76825600] via 30.30.30.1, 00:02:35, Vlan30
D 2.0.0.0/24 is subnetted, 3 subnets
D 2.2.1.0 [90/76825600] via 30.30.30.1, 00:02:35, Vlan30
D 2.2.2.0 [90/76825600] via 30.30.30.1, 00:02:35, Vlan30
D 2.2.3.0 [90/76825600] via 30.30.30.1, 00:02:35, Vlan30
D 3.0.0.0/24 is subnetted, 3 subnets
C 3.3.1.0 is directly connected, Vlan7
C 3.3.2.0 is directly connected, Vlan8
C 3.3.3.0 is directly connected, Vlan9
D 10.0.0.0/24 is subnetted, 1 subnets
D 10.10.10.0 [90/51225600] via 30.30.30.1, 00:02:36, Vlan30
D 20.0.0.0/24 is subnetted, 1 subnets
D 20.20.20.0 [90/51225600] via 30.30.30.1, 00:02:36, Vlan30
D 30.0.0.0/24 is subnetted, 1 subnets
C 30.30.30.0 is directly connected, Vlan30